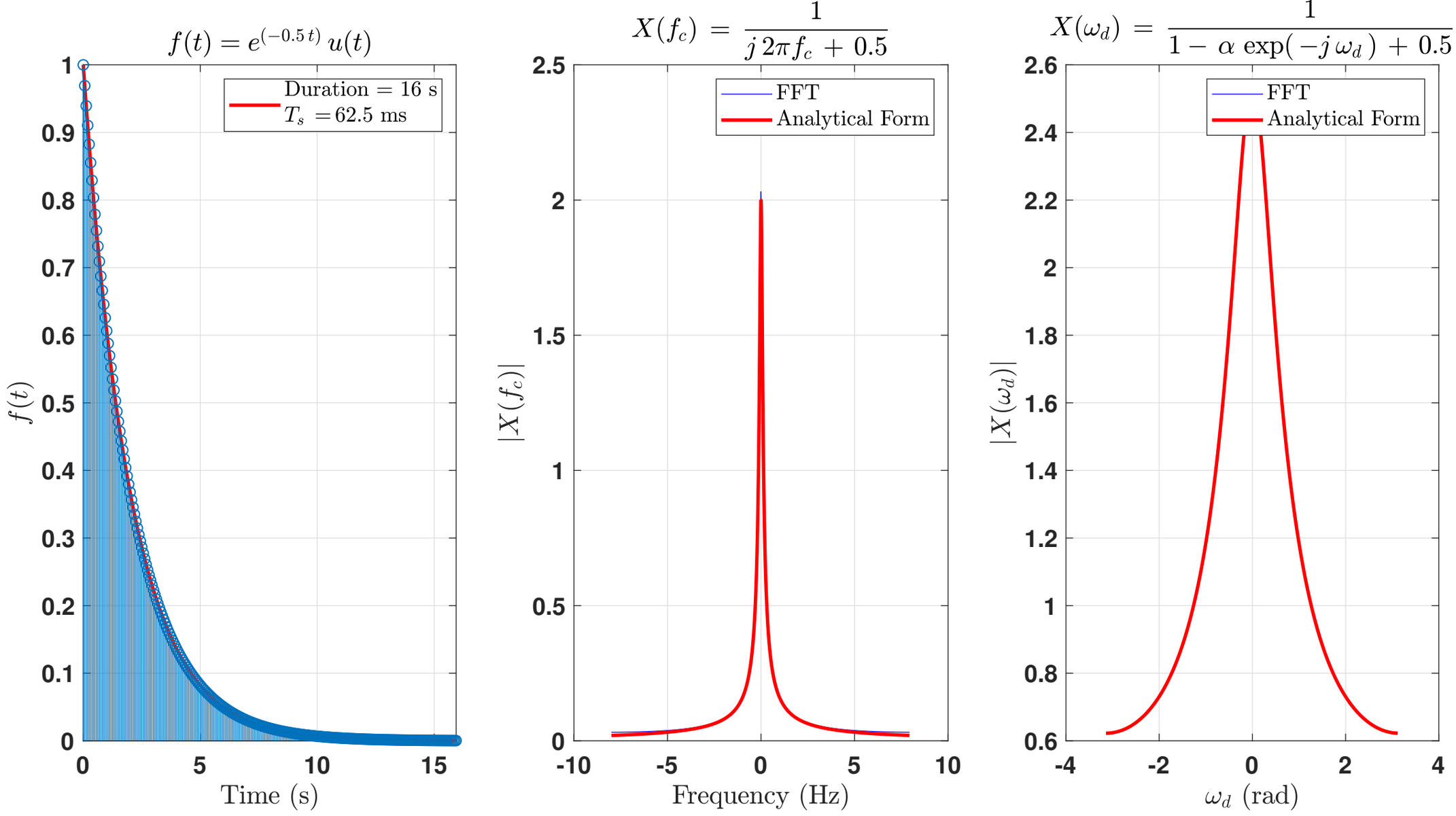

$f(t) = e^{(-0.5\,t)}\,u(t)$
Duration = 16 s
T_s = 62.5 ms
$f(t)$
Time (s)
$X(f_c) = \frac{1}{j\,2\pi f_c + 0.5}$
FFT
Analytical Form
$|X(f_c)|$
Frequency (Hz)
$X(\omega_d) = \frac{1}{1 - \alpha\,\exp(-j\,\omega_d) + 0.5}$
FFT
Analytical Form
$|X(\omega_d)|$
ω_d (rad)